

Luca Luchini

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EDUCATION

OAKLAND UNIVERSITY — B.S. Mechanical Engineering

Rochester Hills, MI

Expected May 2028

EXPERIENCE

BGM ENGINEERING

Shelby Twp, MI

Mechanical Engineering Intern

May 2026 – Present

- Designed and developed production fixtures in SolidWorks, taking concepts from CAD through prototyping, fabrication, assembly, and implementation on the production floor.
- Used Fusion 360 CAM and CNC machining to manufacture fixture components and perform precision rework on production hardware, including CNC-drilled modifications to electronic assemblies.
- Rapidly prototyped and iterated fixture designs using FDM 3D printing, refining geometry based on fit, usability, manufacturability, and production feedback.
- Created engineering drawings and manufacturing documentation for custom components, including parts produced by external machine shops.
- Worked alongside mechanical, electrical, manufacturing, and production personnel to troubleshoot hardware issues and develop practical improvements to assembly and test processes.

Summer Intern

May 2025 – Aug 2025

- Designed a load-cell monitoring system to measure conformal-coat tank levels, preventing downtime and avoiding a commercial system costing several thousand dollars.
- Streamlined assembly and test-line workflow by improving process steps and implementing practical efficiencies to increase throughput.
- Gained exposure to the product-development process from design through final assembly, including design challenges, manufacturing constraints, and practical implementation.

WMU Formula SAE — Electrical & Power Delivery

Sep 2024 – May 2026

Kalamazoo, MI

- Contributed to redesign of the rear differential mounting system by identifying failure points and refining geometry to improve structural reliability.
- Modeled drivetrain components in SolidWorks and used FEA to evaluate stress distribution, factor of safety, and expected loading.
- Supported EV electrical and power-delivery work, gaining experience with drivetrain packaging, electrical integration, and system-level design.

BELFOR USA — 3D Scan Technician

May 2022 – May 2025

Birmingham, MI

- Provided 3D building scanning and modeling services using Matterport for residential and commercial clients across the U.S.

SELECTED PROJECTS [View Full Portfolio](#)

PCB Test & Programming Fixture

SolidWorks | Fusion 360 | CNC | 3D Printing

- Designed and built a compact fixture for repeatable electrical contact with densely spaced PCB test points.
- Improved positional accuracy by replacing critical FDM features with a CNC-machined Delrin insert and integrating pogo-pin hardware and wiring.

Optical Measurement Fixture System

SolidWorks | Fusion 360 | CNC | 3D Printing

- Designed interchangeable fixtures for repeatable positioning of small electronic assemblies on an optical measurement system.
- Reduced typical board-change downtime from ~20 s to ~5 s, a 75% reduction, using a standardized machine-side locating interface.

TECHNICAL SKILLS

CAD / CAE: SolidWorks, Fusion 360, AutoCAD, FEA

Manufacturing: CNC machining, CAM, 3D printing, GD&T, DFM, metrology

Electronics / Test: KiCad, LTspice, oscilloscopes, multimeters, soldering

Programming: Python, C/C++, MATLAB, Git